

Looking back on the Komodo dragon containment project, there are several things I would recommend to anyone attempting a similar design. There were a lot of things we learned the hard way, and I think future teams could save a lot of time and effort by building on what worked and avoiding some of our mistakes.

First, I'd suggest starting with more detailed research and stakeholder input right from the beginning. We did talk to the client and got their priorities, but if I could go back, I would have made sure to get feedback from more people—like other zookeepers or maintenance staff—who actually work with the animals and the equipment. That kind of input could help avoid issues later, especially with how the doors function or how easy the system is to use in real life.

Another thing I'd recommend is making the design modular from the start. We ended up realizing that being able to swap out parts or make quick adjustments was really helpful, especially when something didn't work exactly as expected. If the pieces were all designed with the same measurements and could be reconfigured easily, it would have made testing and revising a lot faster and more accurate.

One area where we struggled a bit was with the materials. Most of our testing was done using cardboard or lower-strength materials, which gave us an idea of how things might work, but didn't give us the full picture of how the final design would hold up. In the future, I think it would help to either build at least part of the prototype out of the actual materials early on or use simulation software to estimate how things like stress or friction would affect the design over time.

Time management is another thing I'd stress. We were pretty tight on time near the end, and it definitely limited how much testing we could do and how many changes we could make. I'd recommend setting aside at least a couple of weeks just for testing and refining. Even better, having a checklist of specific things to test—like how many times the door can open and close before it starts sticking—would make the process more organized.

I also learned that materials don't always behave the way you expect. For example, we thought the door would slide fine, but it ended up warping a little and getting stuck. In the future, I would build in some extra tolerance or adjust the design to allow for small changes in size or shape due to wear or stress.

Documentation is something I didn't think would be that important at the start, but it ended up making a big difference. Keeping track of what changes we made, what worked and didn't, and having photos or drawings really helped us when we had to go back and fix something. If I were to do this again, I'd make it a point to document everything regularly, maybe even assign someone to keep things organized.

Lastly, I'd say don't be afraid to get creative, even if you're working with limited tools, time, or materials. Some of our best ideas came from working around those limits. At the same time, be realistic about what you can and can't test in the prototype, and make sure to explain those limitations clearly if you're handing the project off to someone else.

To put all these ideas into action, I think a future team could benefit from a more structured schedule. Start with a couple of weeks of solid research and planning, then build and test individual components before putting the full system together. Leave enough time at the end to test everything thoroughly and make changes based on what you learn.

Overall, this project taught me a lot, and I think with better planning, more realistic materials and budgeting, and more time for testing, future versions of this system could be even more effective and easier to use.